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A stroke-based method for automated generation of schematic network maps

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This article deals with the graphic simplification of a network by schematization. A new method employing a stroke-based and progressive strategy is proposed to generate schematic network maps. This method treats a stroke (which is a long line with segments concatenated together) as a basic unit for the implementation. The procedure is as follows: (a) strokes are formed from line segments, (b) the strokes are re-orientated along grid lines and/or diagonals, and (c) two endpoints and all intersection points on (sub-)strokes are projected onto re-oriented straight lines, and (d) spatial inconsistency is detected and resolved. A methodology for each of these steps is described. This new method has been tested with a set of real-life road network data and evaluated by fractal analysis and empirical study. Experimental results show that this new method is more effective than segment-based methods and is able to produce graphics with great simplicity and clarity. Based on the results obtained, the stroke-based schematization with four primary directions is recommended.

Keywords: schematic maps; progressive generalization; stroke; fractal analysis

1. Introduction

Quite often, map producers emphasize the accuracy and the completeness of map content, which are two of the most important elements of map quality. However, in many applications, users are not very satisfied with such maps. Then a question arising is ‘what matters?’ Alternatively, one may ask: ‘Is accuracy (or quality) the most important consideration for users?’ The answer is ‘not necessarily’. Recently, there has been a shift of attention from quality to usability, which is a feature that lays emphasis on needs, abilities and perception skills of potential users (Opach 2010). For example, the Association of Geographic Information Laboratories of Europe (AGILE) has recently established a working group on usability of spatial data (<http://www.modulobus.org/agile-sdu/default.html>), and this working group has organized some workshops and themed issues on usability (e.g. Hunter *et al.* 2003). A set of more than 50 indicators has been developed to measure the usability of spatial data. However, according to an investigation conducted by Bregt (2001), to non-professionals, ‘usability’ means ‘simple, clear not complex’ and ‘easily assessable’.

One example of such ‘simple, clear not complex’ maps is the London Tube map, which has been verified as a spectacular example of ‘usable’ representation of transport networks (Forrest 2003, Opach 2010). Such a map is termed a schematic map, or schematic network map as it is the schematic map of a network. Such schematic maps are effective in conveying

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information of limited scope, making them easy to interpret by concentrating on the relevant aspects of the information and omitting unnecessary details based on linear abstractions of networks. They can be used in any scenario in which streams of objects at nodes in a network play a role (such as routes in a transportation system or cartographic schemes for gas, water or electricity mains) (Avelar 2007).

It is well known that the London Tube map is a result of manual design through many generations since the first design by Harry Beck in 1931 and it has been a dream of cartographers to produce such ‘simple, clear not complex’ maps by automated means.

For the automated generation of schematic maps for underground routes, Scott and Rodgers (2005) employed a graph model. In this model, stations were represented by nodes and routes by edges. They used eight criteria (five for positioning labels and three for positioning nodes) for multicriteria optimization using a hill climber algorithm. For the schematization of general road networks, Elroi (1988) defined three steps, viz.

- (a) simplifying lines to basic shapes;
- (b) re-orienting lines along grid lines; and
- (c) enlarging those areas where roads are congested.

Avelar (2007) has made use of the Douglas–Peucker algorithm to remove details of lines and they adjusted the straightened line network using a dynamic iterative algorithm similar to the steepest descent energy minimization algorithm with a set of constraints (both geometric and aesthetic). Cabello *et al.* (2001) have discussed the mathematical issue of schematization of road networks and designed a combinational algorithm for this purpose. However, their algorithm does not guarantee a solution. Recently, Ware *et al.* (2006) and Anand *et al.* (2007) made use of simulated annealing algorithms for the automated production of schematic maps. In the first step, the Douglas–Peucker algorithm is again used for line simplification. Then the vertices of simplified lines are displaced to produce schematization, which is regarded as an optimization process and is achieved using a simulated annealing (SA) algorithm. Seven constraints were considered for the optimization, viz. topological, orientation, length, angle, rotation, clearance and displacement. Dong *et al.* (2008) have recently reported their work on the production of schematic maps based on an adaptive regular square grid model according to multiple criteria.

Schematic maps produced by existing algorithms are based on road segments. The resulting maps are still very close to the original map. It is very clear that these maps are far too ‘realistic’ when compared to the London Tube map. Indeed, in this article, the authors will present a new method, which is able to automatically generate schematic maps with an abstract level similar to that of the London Tube map.

The rest of this article is organized as follows. In Section 2, a new strategy for schematization is proposed. In Sections 3–6, methodologies for these steps are presented and a description of the prototype implementation is described. In Section 7, a set of real-life data is used to analyze and evaluate the adequacy of the new method. In Section 8, a discussion on multiscale representation is conducted and in Section 9 some conclusions are presented.

2. A new strategy for schematization

Before discussing any strategy for schematization, it is essential to keep some basic principles in mind. Based on the work of Barkowsky and Freksa (1997), a set of properties in hierarchy is considered in the cognition of maps (see Figure 1), i.e. existence and connectedness, orientation, location, distance and shape. However, by experience one also knows that it is the relativity in orientation, location, distance and shape that matters in

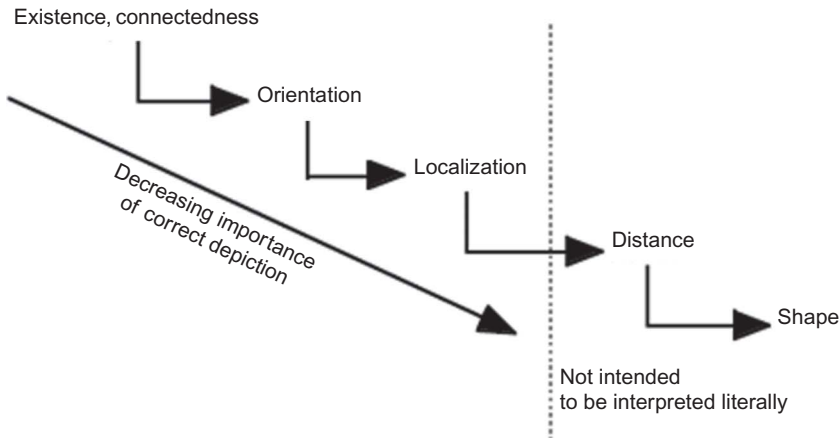


Figure 1. Hierarchy of constraints for cognitive purpose (Barkowsky and Freksa 1997).

spatial cognition. As shape is of least importance among these properties, the shape of a network is to be distorted in the generation of schematic maps in exchange for the clarity of representation and other properties, i.e. the connectedness (or connectivity), relative orientation, relative location (or position) and relative distance, which are to be preserved as much as possible, with the hierarchy in mind.

From a practical point of view, it has been found that some schematic maps generated by existing methods consist of lines with a step-like appearance. Such a representation looks untidy and crowded, leading to unpleasant-looking outcomes. This must be due to the fact that individual line segments were considered separately. If the road segments are concatenated into long lines, then one could orientate the lines into a horizontal/vertical/diagonal aspect. In this way, the result will be clearer and/or more pleasing. Such a concatenated long line has been referred to as a 'stroke' in the literature of geo-information science (Thompson and Richardson 1999), although the term 'stroke' has quite different meanings in other disciplines, such as in language and medicine.

The authors believe that stroke (instead of segment) should be used as the basic unit in schematization. With this line of thought, a new strategy is proposed as follows:

- (a) to form strokes from line segments;
- (b) to re-orientate the strokes along grid lines and/or diagonals;
- (c) to straighten the lines by projection; and
- (d) to detect and resolve spatial inconsistency, if any.

To guarantee that the result from every step satisfies the requirements described above, a 'progressive' strategy is adopted to generate schematic network maps by handling strokes one by one according to the hierarchical structure of the strokes. General methodologies for these steps will be presented in the coming sections.

3. Stroke as basic unit for schematization

In the case when road segments are short, they may be concatenated into a longer line, called a stroke in geo-information science. Named streets and/or named roads are well-identified strokes. That is, the street/road segments with an identical name are sequentially connected together to form a stroke. On the other hand, if streets/roads are not named in a set of data, then

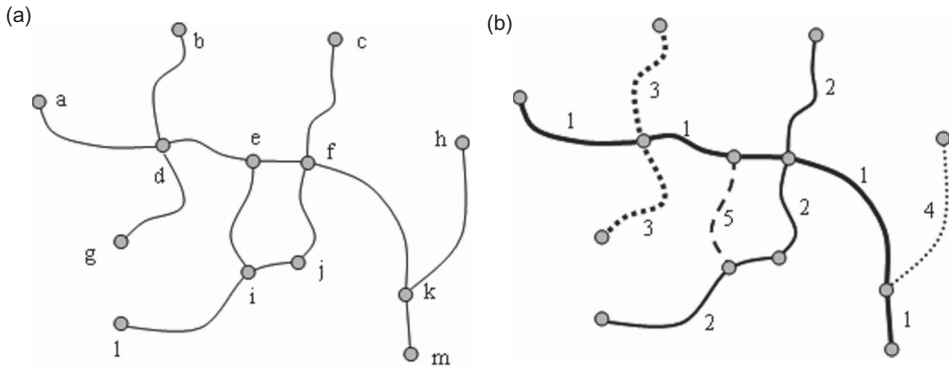


Figure 2. Formation of strokes based on good continuation grouping principle of perception: (a) a road network; (b) segments concatenated into strokes.

one needs to make segments concatenated naturally based on some principles. The Gestalt ‘principle of good continuation grouping’ of perception has been widely used as a theoretical basis for stroke formation in practice (Thompson and Richardson 1999), which says that graphic elements that suggest a continued visual line will tend to be grouped together (Bastoky 2010). Figure 2 is an illustration of concatenating 13 road segments into five strokes.

The natural strokes can be formulated with the following procedure:

- Set a deflection angle threshold. A value 45° is selected in this study as (1) it has been concluded that a deflection angle threshold between 30° and 75° will produce stable outcomes (Jiang *et al.* 2008) and (2) we make use of four and eight directions for re-orientation of lines at a later stage.
- For each line segment, compute all the deflection angles formed by itself and each of all the segments that intersect at the same point (node).
- Form a stroke with the given segment if none of these deflection angles is smaller than the threshold; or else add those segments with deflection angles smaller than the threshold into the set of so-called possible joining segments (PJSs) for consideration.
- Select a segment from the PJS based on one of the three principles of every-best-fit, self-best-fit and self-fit.

The self-fit is the laziest strategy as it just randomly selects a segment from PJS as the next one. Figure 3a shows the case. Here, any of the other three segments (‘b’, ‘c’ and ‘d’, but not ‘e’) can be chosen as the next segment for ‘a’ if the deflection angle is less than a certain threshold. In self-best-fit, each segment is very selfish. It only considers the benefit to itself of choosing a segment from PJS where the deflection angle with itself is the smallest. In this case, ‘a’ is connected to ‘b’ (see Figure 3b). In every-best-fit, both the given segment and all the PJSs are considered to find the best way to form strokes. In other words, every pair of segments at a junction point has to negotiate with each other to have the best fit. In the case shown in Figure 3c, although segments ‘a’ and ‘b’ have the minimum deflection angle if viewed from segment ‘a’, yet segments ‘b’ and ‘e’ have a smaller deflection angle if viewed from segment ‘b’ or ‘e’, and thus segment ‘a’ will be connected to ‘c’ so as to let segments ‘b’ and ‘e’ have the priority to connect each other (Figure 3c).

Among these three, every-best-fit generates a unique stroke network for a given road network. This is the method used by Thompson and Richardson (1999) and is most widely

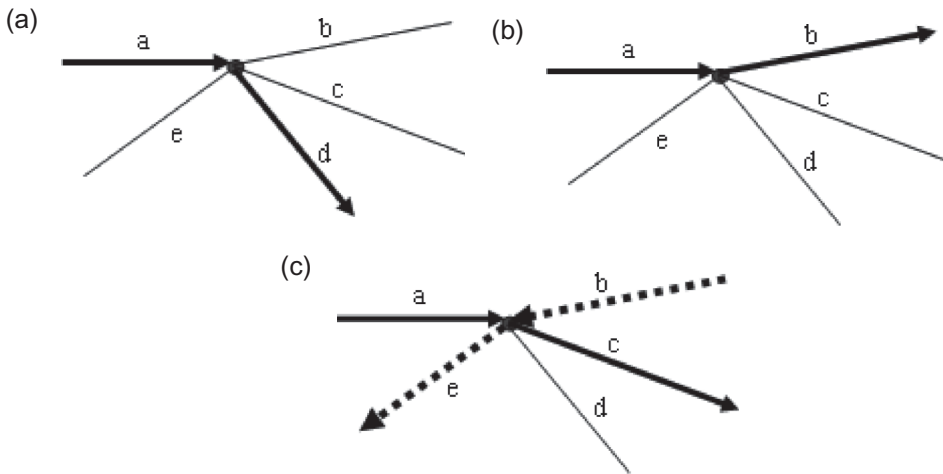


Figure 3. Three types of method for selecting a segment to connect: (a) self-fit; (b) self-best-fit; (c) every-best-fit.

used in practice. More detailed information on these three methods can be found in a recent paper by Jiang *et al.* (2008).

It has also been discussed (Heinzle and Anders 2007, Li 2007, Chen 2009) that attributes of road segments (such as road type, width, length and number of lanes) and of road intersections (e.g. degree of connectivity and characteristics of intersections) can also be used as weights for road segments in stroke formation. Furthermore, strokes can also be organized into a hierarchy for multiscale representation (see discussion in Section 8).

4. Two options for re-orientation of strokes

It is understandable that the step-like appearance of the schematic maps generated by existing algorithms is mainly due to the fact that the orientated and straightened segments are too short so that a long line may have too many differently oriented segments. Therefore, the orientation of a long stroke is more complex than that for a short segment and is worthy of discussion here.

First, one should consider the number of directions to which the strokes are oriented; there are two options in general:

- four primary directions (i.e. along grid lines as priority), and
- eight directions (i.e. along grid lines and diagonals with equal opportunity).

Figure 4 shows an example of these two options. In reality, it might not be possible to re-orientate all lines to only the four primary directions without creating congestions. That is, by four primary directions, it is meant that the vertical and horizontal directions are given a high priority and the diagonal directions are considered as a secondary choice. On the other hand, in the case of eight directions, all directions are given equal opportunity.

Second, one needs to check whether or not a long stroke can be re-orientated to a single direction. If not, the stroke will be bent to fit into more than one direction. For example, Stroke 2 in Figure 4a has been fitted into three directions, according to the trend. Two approaches can be used to check such a necessity:

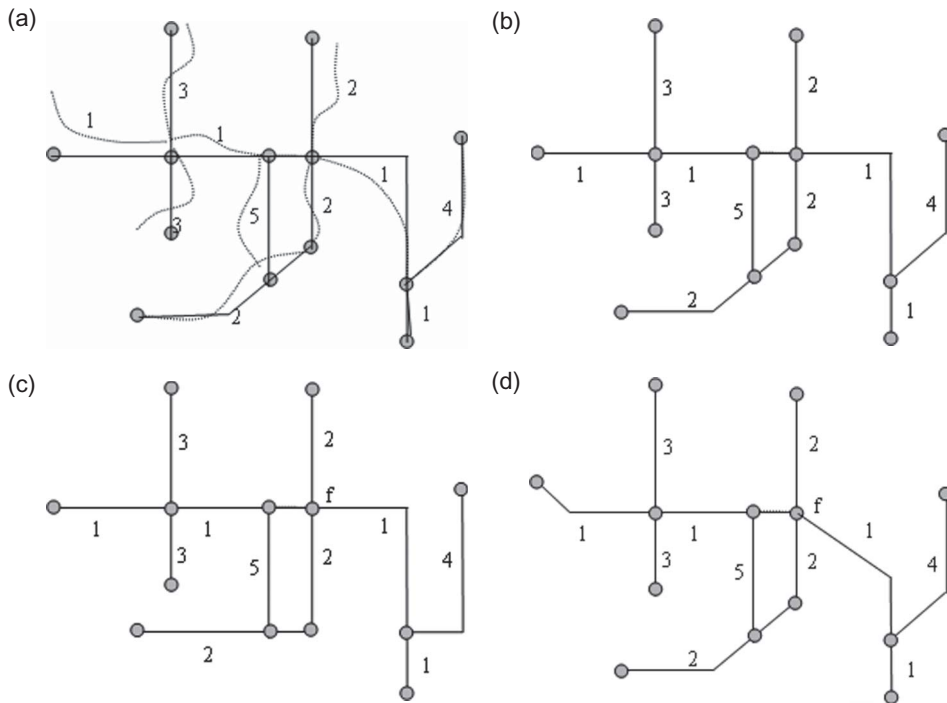


Figure 4. Two options for re-orientation of strokes. (a) Orientate and straighten strokes. (b) Possible result 1: fitted and straightened. (c) Possible result 2: with four primary directions. (d) Possible result 3: with eight directions.

- The first approach – maximum direction distortion checking: to check the possible maximum direction distortions against a given threshold (e.g. 45° according to the four primary directions mentioned above), which are to be introduced if the straight line connecting the first and last point of the stroke is used to represent the stroke itself. If no direction distortion is larger than a given threshold, then the stroke is straightened and oriented accordingly. Otherwise, two orientations should be considered.
- The second approach – sequential direction change checking: to check the direction changes sequentially. If a change in direction at a point is larger than a threshold, then a change of orientation at this point is warranted.

The working procedure for the first approach is similar to the Douglas–Peucker algorithm (Douglas and Peucker 1973). But in this case, the direction distortion, instead of the perpendicular distance, will be considered. Figure 5a shows how the direction distortion is used as a criterion to split a stroke. If line $\overline{a, m}$ is used to represent the stroke, there will be two direction distortions for each point in between a and m . If the largest direction distortion is still smaller than a threshold, then $\overline{a, m}$ is a good representation of the stroke. On the other hand, the point at which the largest direction distortion is caused will be selected for splitting the stroke into two sub-strokes. ‘N’ in Figure 5a is such a point. The subdivision process will continue until the direction distortion is smaller than the threshold. Figure 5b shows the working of the second approach. In this case, abrupt changes in direction at points ‘f’ and ‘k’ are found and thus this stroke will be re-orientated into three directions.

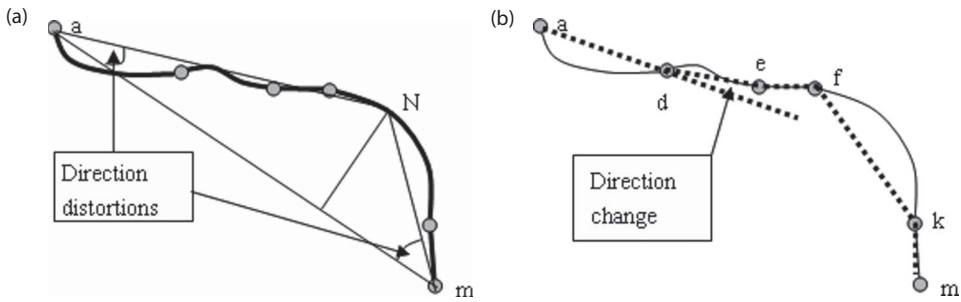


Figure 5. Checking the necessity of re-orientating a stroke into more than one direction: (a) maximum distortion; (b) sequential direction change.

5. Two types of projections for mapping points on a sub-stroke onto a straight line

After the re-orientation of a sub-stroke is determined, the two endpoints of all intersection points and all intermediate points on this sub-stroke are then projected onto the straight line. As a result, original segment orientations of this stroke or sub-stroke will re-orientate to the same direction. The two endpoints of all intersection points will be projected and retained, while all intermediate points will be removed to make the result simpler and clearer, which in essence omits the line simplification process required in segment-based methods. Two types of projection (Figure 6) can be used:

- perpendicular projection and
- proportional projection.

The first is to project points perpendicularly onto the straight line (Figure 6). Figure 6a shows such a projection. In this way, all new points will have their original X -coordinates kept unchanged but their Y -coordinates changed into an identical value. Mathematically,

$$\begin{aligned} X_{i,\text{new}} &= X_{i,\text{old}} \\ Y_{i,\text{new}} &= Y_{\text{common}} \end{aligned} \quad (1)$$

where Y_{common} is the Y -coordinate common to all the points on this sub-stroke. The determination of its value is worthy of some discussion. The main consideration should be the Y -coordinates of the intersections of the strokes. For example, the point 'f' in Figure 5 is the intersection of two major strokes (strokes 1 and 2); then the Y -coordinate of this point should be given higher weight than other points on the sub-stroke from point 'a' to 'N' in Figure 5a.

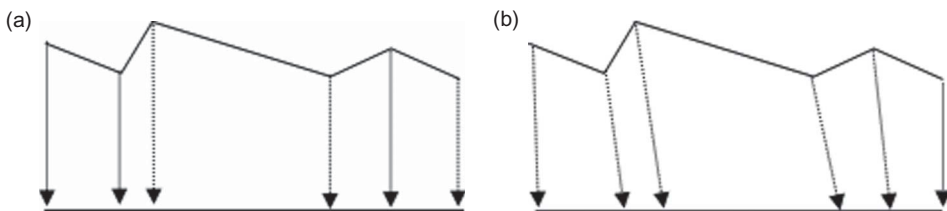


Figure 6. Two types of projections for mapping points on a sub-stroke onto a re-oriented line: (a) perpendicularly projected; (b) proportionally projected.

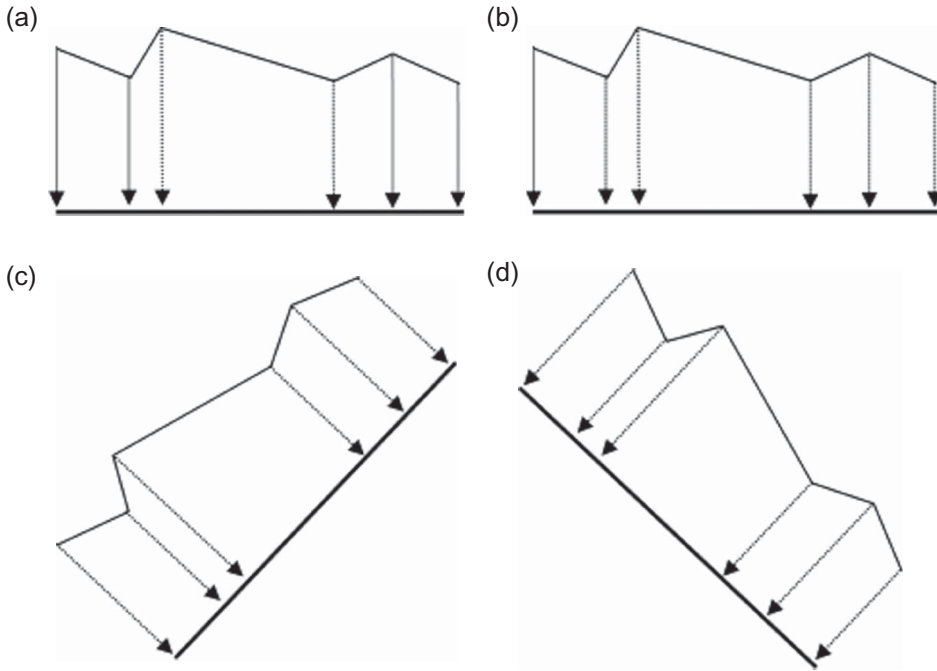


Figure 7. Four different cases in vertical projection: (a) projected to a horizontal line; (b) projected to a vertical line; (c) projected to a 45° diagonal; (d) projected to a -45° diagonal.

This is the case when a sub-stroke is projected to a horizontal line. However, there are also three other cases (Figure 7). In the case of projection onto a vertical line, the equations become

$$\begin{aligned} Y_{i,\text{new}} &= Y_{i,\text{old}} \\ X_{i,\text{new}} &= X_{\text{common}} \end{aligned} \quad (2)$$

In the case of projection onto a diagonal line, a three-step procedure can be applied, i.e. (a) to rotate the sub-stroke by 45° (or -45°); (b) to project points on the sub-stroke onto the horizontal (or vertical) grid line using Equation (1) (or Equation (2)); and (c) transforming the projected coordinates back to the original coordinate system by inverse rotation.

Perpendicular projection will guarantee the preservation of relativity in position. If the direction changes at the points on the sub-stroke are not large, this method will work nicely and simply. This should be the normal case because if the direction change at a point is larger than a threshold, then two sub-strokes are formed and they will be re-oriented into different directions.

By proportional projection, the two endpoints and all intersection points on this sub-stroke are relocated along the straight line with intervals proportional to the lengths on the sub-stroke (see Figure 6b). However, this method was not used in our system because the perpendicular projection is able to preserve the relativity in length in this stroke-based approach.

6. Hierarchical structure of strokes for progressive schematization

As discussed in Sections 4 and 5, it is straightforward to re-orientate an individual (sub-)stroke onto a given direction (among the four grid and four diagonal directions) and to project points onto the direction. However, the re-orientation and projection may cause spatial conflicts

(mainly topological inconsistency) between neighboring strokes. Therefore, a strategy is required to guarantee topological consistency.

In this study, a ‘progressive’ strategy is adopted to generate schematic network maps. ‘Progressive’ means that strokes are handled one by one in the schematization process and, in the handling of the current stroke, the previously schematized strokes are treated as references. This guarantees that the results from each step for each stroke are reasonable. To achieve this, the strokes are put into an order according to their priorities (or rankings). The ranking values are computed by Equation (3):

$$S_w = \begin{cases} S_t & \text{if } S_{t1} \neq S_{t2} \\ S_l & \text{if } S_{t1} \equiv S_{t2} \cap S_{l1} \neq S_{l2} \\ S_d & \text{if } S_{t1} \equiv S_{t2} \cap S_{l1} = S_{l2} \cap S_{d1} \neq S_{d2} \end{cases} \quad (3)$$

where S_t is the stroke type, S_l is the length and S_d is the stroke degree (i.e. number of intersections).

That is, in the ranking, type is taken as the first consideration, then the length, and finally the degree of the stroke. Figure 8 shows the hierarchical structure of the road network shown in Figure 2b. In the re-orientation and projection, Stroke 1, which has the highest ranking, is handled first. The neighboring strokes will be examined to see whether there is any inconsistency created, as shown in Figure 9. If yes, the inconsistency is resolved first. Then, Stroke 2 will be handled and the others follow.

In this study, the point-in-polygon solution is used to detect possible inconsistency. The polygon is formed by the (sub-)stroke and the projected (straightened) stroke. If no points from other strokes are in the polygon, then no spatial inconsistency is created. On the other hand, if one or more points do exist, then we need to move the point(s) out of the polygon. Figure 9c and e show such cases and Figure 9d and f show the resolutions for these two cases, respectively.

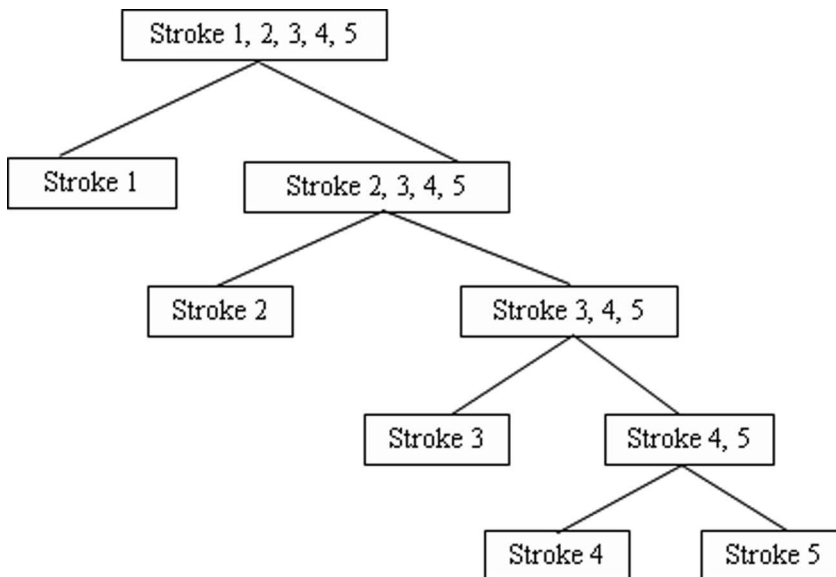


Figure 8. Hierarchy structure of road network based on strokes.

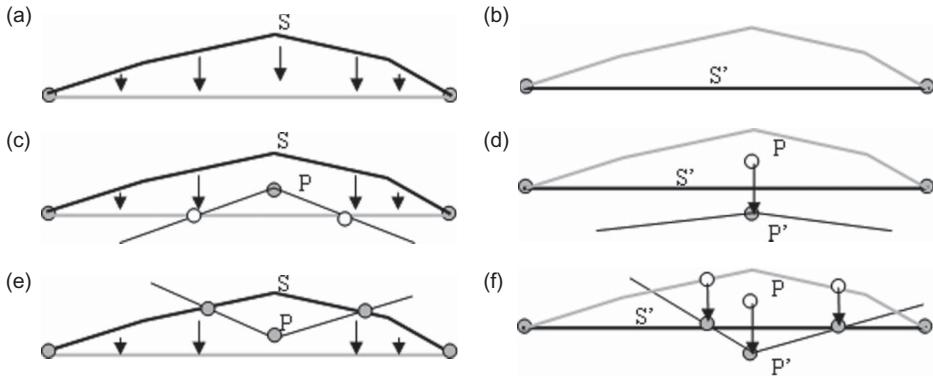


Figure 9. Two cases introduce topological inconsistency. (a) Projection to a horizontal line. (b) No point in polygon. (c) Intersection created after projection. (d) Inconsistency resolution for (c). (e) Intersections lost after projection. (f) Inconsistency resolution for (e).

The whole schematization algorithm can be written as follows:

Input: Segment-based original road lines L ; deflection angle threshold T ;

Output: Stroke-based schematized road lines S ;

Main Function ()

{

FormStrokes (L , T);

ProgressiveSchematization (S);

Sub Function FormStrokes (L , T)

{

 Get all segments with names from L ;

While (there is segment not processed) **do**

 Get a segment as start segment to be processed;

If (this and another segment with an identical name and sequentially adjacent) **then**

 Join these two segments into a new segment;

 Change the status of that segment to 'having been processed';

End If

 Create a new stroke between these segments and add to stroke-based road lines S ;

End While

 Get all segments with no names from L ;

While (there is segment not processed) **do**

 Get a segment as start segment to be processed;

If (deflection angle between this and adjacent segment $> T$) **then**

Return new segment = old segment;

Else

If (every-best-fit with an adjacent segment found) **then**

 Join these two segments into a new segment;

 Change the status of that segment having been processed;

End If

End If

 Create a new stroke between these segments and add to stroke-based road lines S ;

End While

```

}
Sub Function ProgressiveSchematization(S)
{
  Get all strokes from S;
  Rank the strokes from high type to low type according to the strokes weight  $S_w$ ;
  If (having same type) then
    Strokes rank according to length from long to short;
  Else If (having same type and length) then
    Strokes rank according to the degree of the stroke (i.e. number of segments) from
    high to low;
  While (there is stroke not processed) do
    Get a rank highest stroke as start stroke to be schematized;
    Compute and select vertex with maximum direction distortion of this stroke;
    If (maximum direction distortion > T) then
      Split this stroke into two sub-strokes;
    Else
      Compute orientation of the stroke or these two sub-strokes;
      Form a polygon between this stroke and the grid line;
      If (there is any point in the polygon) then
        Move this point out the polygon to the new location;
      Else
        Project two endpoints, junction points and intermediate points onto
        appropriate grid line;
        Change the status of that stroke having been processed;
      End If
    Get next rank highest stroke which not processed according to the stroke weight;
  End While
}

```

7. Experimental evaluations and discussions

In order to validate the applicability of this stroke-based method for automated schematization of a road network, a set of real-life transportation network data are used for testing. The original data and the results are shown in Figure 10. The strokes are the named roads in this case. These strokes are then ranked based on the formula expressed in Equation (3). A three-level hierarchy is built for the set of roads as shown in Figure 10, where the thicker lines represent the higher ranking. The results from this stroked-based method are shown in Figures 12 and 13. To make a comparison, the result obtained from the segment-based method is also presented in Figure 11.

To make a relative comparison, two approaches are used, i.e. mathematical analysis and empirical study. In mathematical analysis, fractal dimension is used as a measure.

7.1. Fractal analysis for schematic network maps

The complexity of a road network can be reflected by fractal dimension because the fracture structure has characteristics of invariability of scale and self-similarity. It has been found that the fractal dimension of the original road network is 1.155. The fractal dimensions for the results are 1.106, 1.099 and 1.096, for the segment-based, stroke-based with eight directions

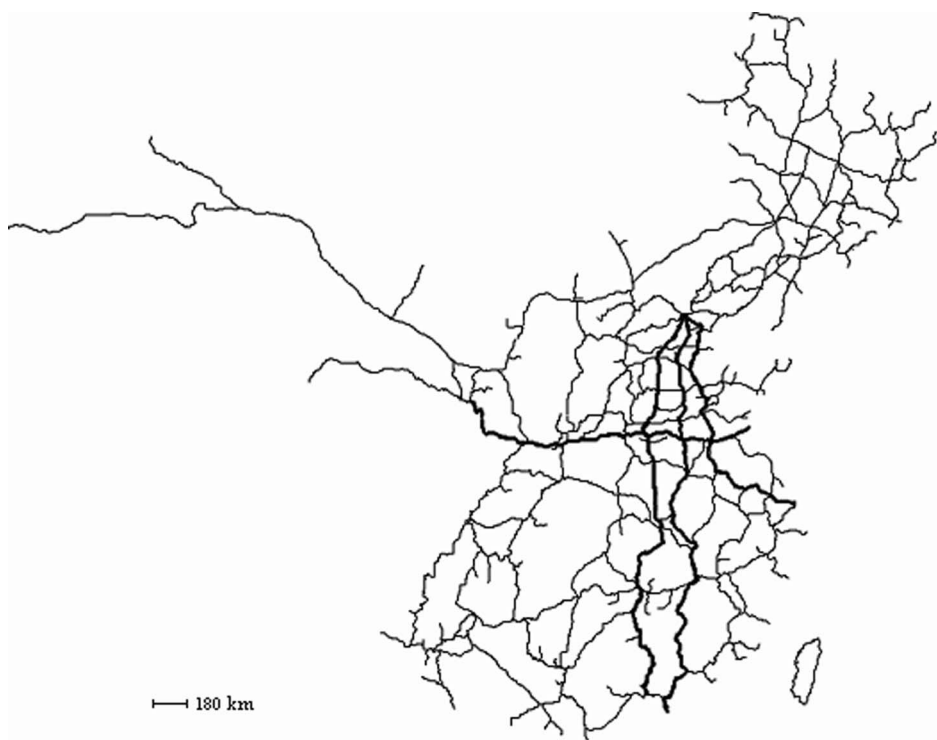


Figure 10. Original network.

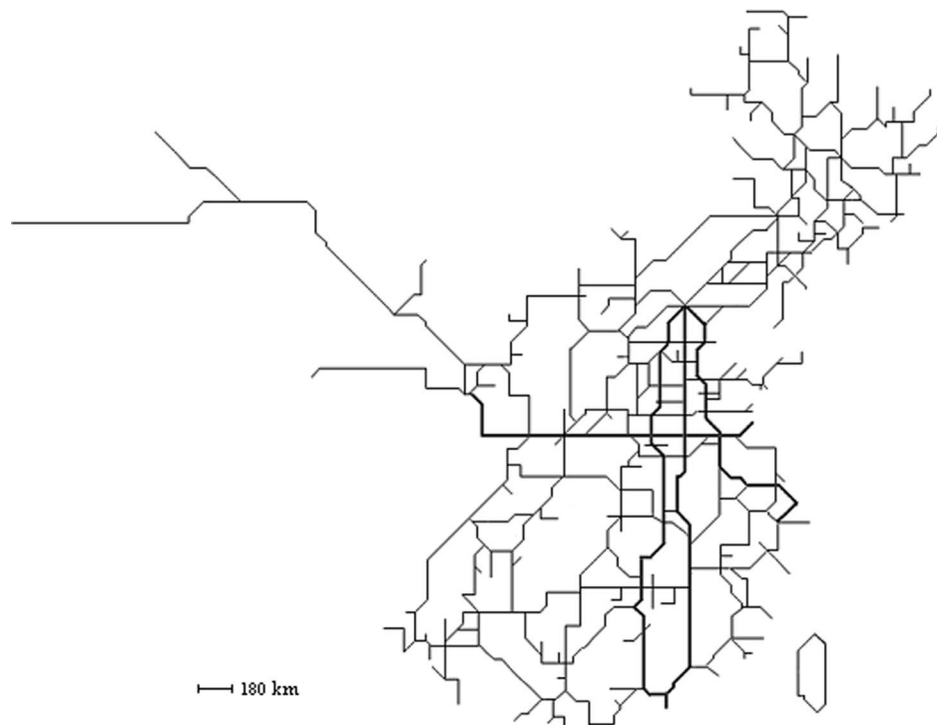


Figure 11. Segment-based schematization result.

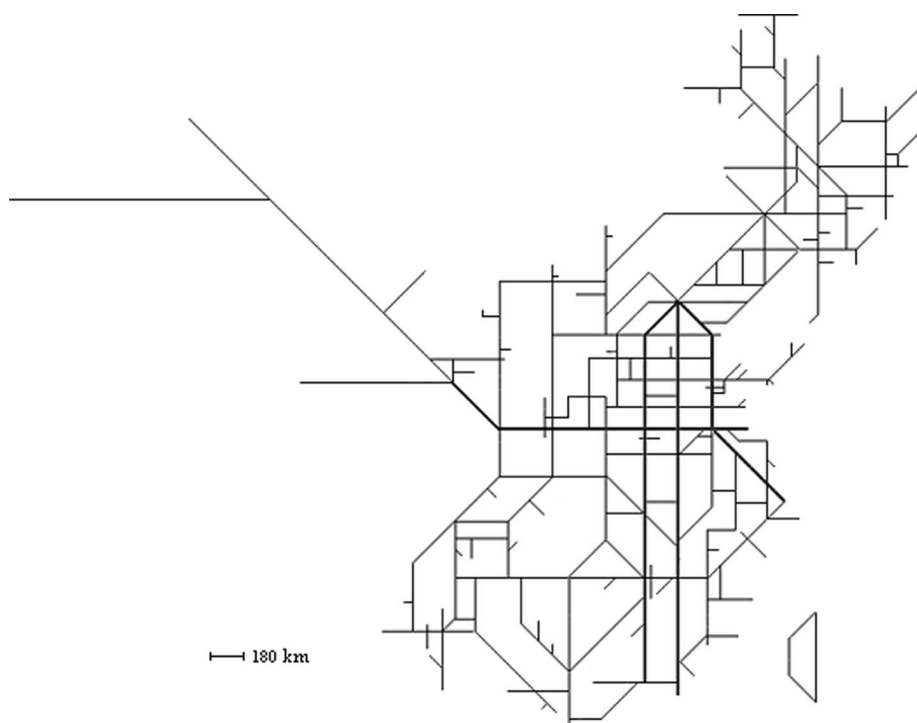


Figure 12. Stroke-based schematization result, with eight directions.

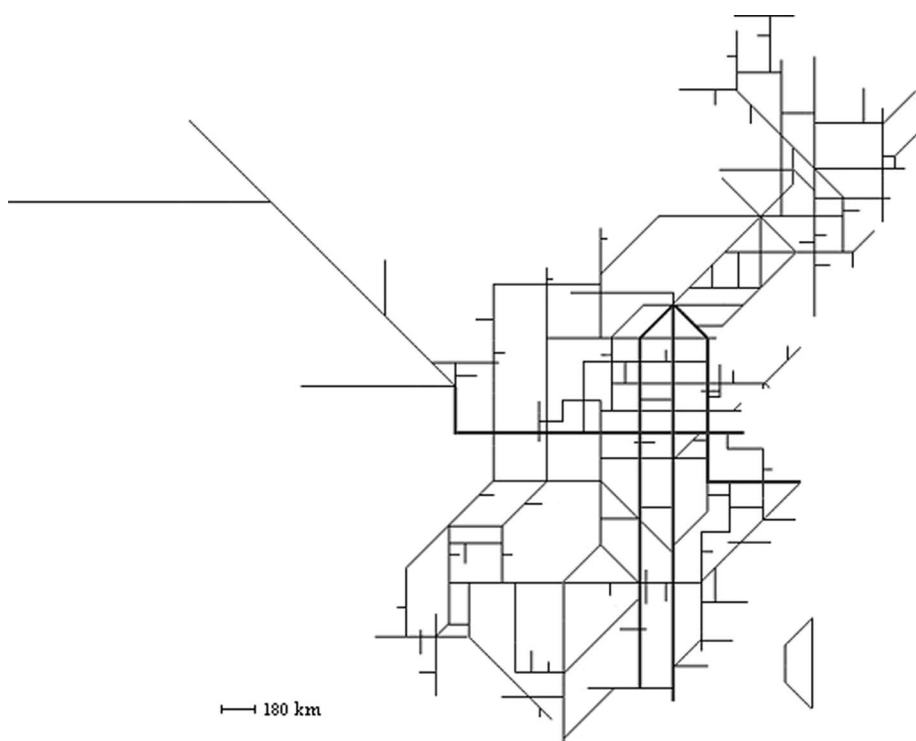


Figure 13. Stroke-based result, with four primary directions.

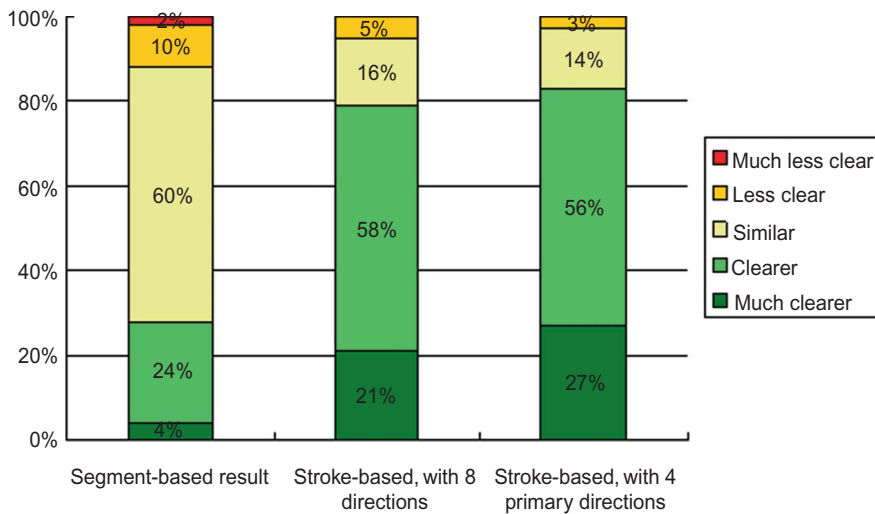


Figure 14. Empirical studies for evaluation of schematic maps.

and stroke-based with four primary directions methods, respectively. The results indicate that the stroke-based method produces results with greater simplicity and clarity.

7.2. Empirical study

It is noted here that the differences in fractal dimension are not very large. We would like to confirm whether human readers also have similar views or not. In this study, a psychological test is carried out, with 104 (54 female and 50 male) students with geography backgrounds as subjects. They were given the maps in Figures 10–13 and asked to give a score to each of these schematic maps by comparison with the original maps. A 5-grade marking system is used as follows: 1 = much less clear, 2 = less clear, 3 = similar, 4 = clearer, 5 = much clearer.

The results are shown in Figure 14. It is interesting to note that only 60% of the subjects think the results from the segment-based method are similar to the original network, whereas only 28% of the subjects think the result is clearer or much clearer than the original one. On the other hand, 79% of the subjects think the result of the stroke-based method with eight directions is clearer or much clearer than the original map while 83% of the subjects think the stroke-based method with four primary directions is clearer or much clearer than the original map. This indicates that the improvement in clarity is huge.

8. Discussion: flexibility for multiscale representation

It must be pointed out here that one of the big advantages of the stroke-based method is the flexibility for multiscale representation of schematic maps. This is because each stroke can be assigned a significance value and strokes can easily be organized into a hierarchy, as discussed in Section 6. With zooming out, strokes with lower significance and at lower levels in the hierarchy would be eliminated gradually. Thus the levels of detail are reduced accordingly. Figure 15 shows results at three different scales based on three levels of hierarchy.

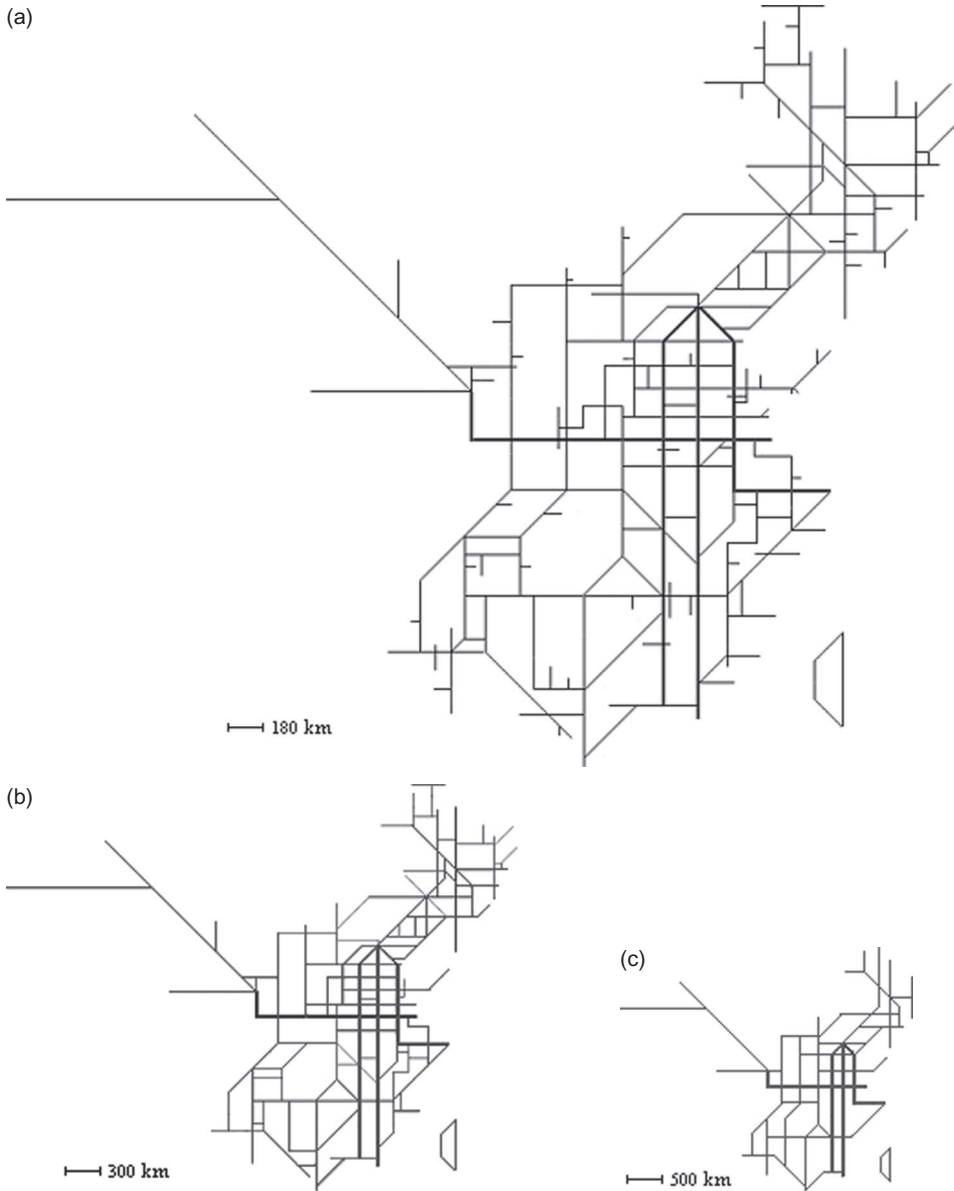


Figure 15. Multiscale representation of schematic map for a transportation network: (a) network with three levels; (b) network with two levels; (c) network with one level.

9. Conclusions

In this article, a new method for automated generation of schematic maps has been proposed. This method treats a stroke (which is a long line with segments concatenated together) as a basic unit. A new strategy has been proposed for implementation of this method, viz. (a) to form strokes from line segments; (b) to re-orientate the strokes along grid lines and/or diagonals; and (c) to project two endpoints and all intersection points on a (sub-) stroke onto the oriented line; and (d) to detect and resolve spatial inconsistency, if any.

As discussed in Sections 3–6, a methodology for each of the four steps has been described. This new method has been tested with a set of real-life road network data. Fractal analysis and empirical study have been employed for evaluation. Both confirm that the results generated by this new method have much greater simplicity and clarity. Therefore, it might be concluded that this stroke-based method sounds well in theory and works well in practice. Based on the results shown in Figures 11–13, stroke-based schematization with four primary directions would be recommended.

With a stroke-based method, the line simplification procedure is not required anymore, because a stroke line is already simplified when the points on it are projected onto a straight line. This is an advantage over segment-based methods, in which the results will be dependent on the simplification algorithm used. Furthermore, with a stroke-based method, it is easy to produce levels of detail of the network for multiscale representation.

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References

- Anand, S., *et al.*, 2007. Automated schematic map production using simulated annealing and gradient descent approaches. *In: Proceedings of the 15th annual GIS research UK conference (GISRUK 2007)*, 11–13 April 2007. Dublin, Ireland (CD-Rom).
- Avelar, S., 2007. Convergence analysis and quality criteria for an iterative schematization of networks. *Geoinformatica*, 11 (4), 497–513.
- Barkowsky, T. and Freksa, C., 1997. Cognitive requirements on making and interpreting maps. *In: S. Hirtle and A. Frank, eds. Spatial information theory: a theoretical basis for GIS*. Berlin: Springer, 347–361.
- Bastoky, G., 2010. Fundamentals of document design for the technical writer. Available from: http://www.bastoky.com/docdesign_home.htm [Accessed 3 March 2010].
- Bregt, A., 2001. Users' perception of spatial data usability. *In: Presentation at Workshop on spatial data usability*, 19–20 November 2001. Delft, The Netherlands.
- Cabello, S., *et al.*, 2001. Schematization of road networks. *In: Diane L. Souvaine Ed., Proceedings of the 17th annual symposium on computational geometry*, 3–5 June 2001. Medford, MA: ACM, 33–39.
- Chen, J., *et al.*, 2009. Selective omission of road features based on mesh density for automatic map generalization. *International Journal of Geographical Information Science*, 23 (8), 1013–1032.
- Dong, W.H., Liu, J.P., and Guo, Q.S., 2008. Visualizing schematic maps through generalization based on adaptive regular square grid model. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XXXVII (Pt. B2), 379–383.
- Douglas, D.H. and Peucker, T.K., 1973. Algorithms for the reduction of the number of points required to represent a digitized line or its caricature. *The Canadian Cartographer*, 10 (2), 112–122.
- Elroi, D.S., 1988. Designing a network line map schematization software enhancement package. *In: Proceedings of the 8th international ESRI user conference*. Redlands, CA.
- Forrest, D., 2003. The top ten maps of the twentieth century: a personal view. *The Cartographic Journal*, 40 (1), 5–15.
- Heinzle, F. and Anders, K.-H., 2007. Characterising space via pattern recognition techniques: identifying patterns in road networks. *In: William A. Mackaness, Anne Ruas and L. Tiina Sanjakoski: Ed., Generalisation of geographic information: cartographic modelling and applications*, Oxford: Elsevier, 233–253.
- Hunter, G., Wachowicz, M., and Bregt, A., 2003. Understanding spatial data usability. *Data Science Journal*, 2, 79–89.

- Jiang, B., Zhao, S., and Yin, J., 2008. Self-organized natural roads for predicting traffic flow: a sensitivity study. *Journal of Statistical Mechanics: Theory and Experiment*. DOI: 10.1088/1742-5468/2008/07/P07008. Available from: <http://www.iop.org/EJ/abstract/1742-5468/2008/07/P07008> (accessed July 2008).
- Li, Z., 2007. *Algorithmic foundation of multi-scale spatial representation*. Boca Raton, FL: CRC Press – Taylor & Francis Group, 280.
- Opach, T., 2010. Cartography and graphic design. In: G. Gartner and F. Ortig, eds. *Cartography in Central and Eastern Europe, Lecture Notes in Geoinformation and Cartography*. Berlin/Heidelberg, Germany: Springer-Verlag, 199–210.
- Scott, J.M. and Rodgers, P., 2005. Automatic metro map design techniques. In: *Proceedings of the XXII International Cartographic Conference (ICC2005)*, 11–16 July 2005. A Coruna, Spanish (CD-ROM).
- Thompson, R.C. and Richardson, D.E., 1999. The ‘Good Continuity’ principle of perceptual organization applied to the generalization of road networks. In: *Proceedings of the 19th international cartographic conference*, Ottawa, 1215–1225.
- Ware, J.M., et al., 2006. Automated production of schematic maps for mobile applications. *Transactions in GIS*, 10 (1), 25–42.